

International Classification of Retinopathy of Prematurity, 3rd edition (ICROP3) Classification

Each eye should be classified based on zone, plus disease, stage, and extent. If aggressive ROP (A-ROP) is present, it should be noted.

1. Zone

- a. Definition of 3 retinal zones centered on the optic disc. The location of the most posterior retinal vascularization or ROP lesion denotes the zone for the eye.
- b. Definition of a "posterior zone II" region that begins at the margin between zone I and zone II and extends into zone II for 2 disc diameters.
- c. The term "notch" is used to describe an incursion by the ROP lesion of 1–2 clock hours into a more posterior zone. The ROP zone for such eyes should be noted by the most posterior zone of retinal vascularization with the qualifier "notch" (e.g. "zone I secondary to notch").

2. Plus and Pre-Plus disease: Plus disease is defined by the appearance of dilation and tortuosity of retinal vessels, and pre-plus disease is defined by abnormal vascular dilation and/or tortuosity insufficient for plus disease. Recognition that retinal vascular changes in ROP represent a continuous spectrum from normal to pre-plus to plus disease, with sample images demonstrating this range. These changes should be assessed by vessels within zone I, rather than from only vessels within the field of narrow-angle photographs and rather than from the number of quadrants of abnormality.

3. Stage of acute disease (stages 1–3): defined by appearance of a structure at vascular-avascular juncture as stage 1 (demarcation line), stage 2 (ridge), and stage 3 (extraretinal neovascular proliferation or flat neovascularization). If more than one ROP stage is present in the same eye, the eye is classified by the most severe stage.

4. Aggressive ROP (A-ROP): the term aggressive-posterior ROP (AP-ROP) was previously used to describe a severe, rapidly-progressive form of ROP located in posterior zone I or zone II. Because of increasing recognition that this may occur beyond the posterior retina and in larger preterm infants, particularly in regions of the world with limited resources, the Committee recommends the new term A-ROP.

5. Retinal detachment (Stages 4–5)

- a. Stages of retinal detachment are defined as stage 4 (partial: 4A with fovea attached, 4B with fovea detached) and stage 5 (total).
- b. Definition of stage 5 sub-categories: Stage 5A, in which the optic disc is visible by ophthalmoscopy (suggesting open-funnel detachment); Stage 5B, in which the optic disc is not visible due to retrolental fibrovascular tissue or closed-funnel detachment; and Stage 5C, in which stage 5B is accompanied by

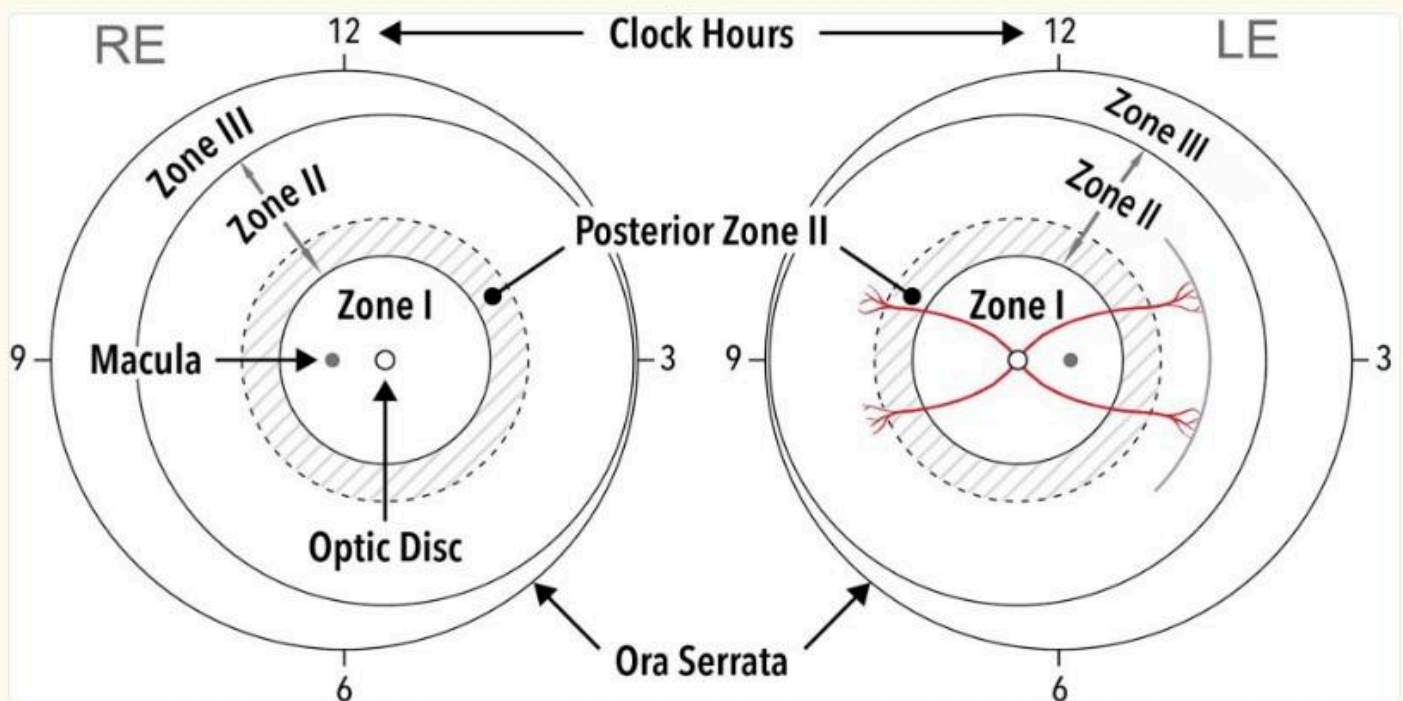
anterior segment changes (e.g. marked anterior chamber shallowing, irido-corneo-lenticular adhesions, corneal opacification), suggesting closed-funnel configuration.

6. **Extent** of disease: defined as 12 sectors in using clock hour designations.

7. **Regression**: definition of ROP regression and its sequelae, whether spontaneous or following laser or anti-vascular endothelial growth factor (anti-VEGF) treatment. Regression can be complete or incomplete. Location and extent of persistent avascular retina (PAR) should be documented.

8. **Reactivation**: definition and description of nomenclature representing ROP reactivation following treatment, which may include new ROP lesions and vascular changes. When reactivation of ROP stages occurs, the modifier “reactivated” (e.g. “reactivated stage 2”) is recommended

9. **Long-term sequelae**: emphasized beyond previous versions of ICROP, including sequelae such as late retinal detachments, PAR, macular anomalies, retinal vascular changes, and glaucoma.



Schema of right eye (RE) and left eye (LE) showing zone borders and clock hour sectors used to describe the location of vascularization and extent of retinopathy.

Solid circles represent borders of zones I-III, and dotted circles represent borders of posterior zone II (two disc diameters beyond zone I). A hypothetical example of examination findings is shown in the left eye, representing approximately three clock hours of stage 1 disease in zone II (note single line on drawing to document presence of stage 1 disease).